

class11

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```
# install.packages("bio3d")
# install.packages("pheatmap")
# install.packages("jsonlite")
```

```
library(bio3d)
library(pheatmap)
library(jsonlite)
```

```
results_dir <- "hivpr_dimer/"
```

```
# Find all PDB model files in the results folder
pdb_files <- list.files(path = results_dir,
                        pattern = "*.pdb",
                        full.names = TRUE)
```

```
# Print the PDB file names
basename(pdb_files)
```

```
[1] "hivpr_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb"
[2] "hivpr_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb"
[3] "hivpr_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "hivpr_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb"
[5] "hivpr_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb"
```

```
# Align and superpose all AlphaFold models
pdbs <- pdbaln(pdb_files,
               fit = TRUE,
               exefile = "msa")
```

Reading PDB files:

```
hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb
hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb
hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb
hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb
.....
```

Extracting sequences

```
pdb/seq: 1   name: hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb
pdb/seq: 2   name: hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb
pdb/seq: 3   name: hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
pdb/seq: 4   name: hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb
pdb/seq: 5   name: hivpr_dimer//hivpr_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb
```

```
# Print alignment summary
pdbs
```

```

1               .               .               .               .               50
[Truncated_Name:1]hivpr_mono PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_mono PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_mono PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_mono PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_mono PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
*****
1               .               .               .               .               50

51               .               .               .               .               99
[Truncated_Name:1]hivpr_mono GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_mono GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_mono GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_mono GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_mono GGFIVKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
*****
51               .               .               .               .               99
```

Call:

```
pdbsaln(files = pdb_files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
5 sequence rows; 99 position columns (99 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

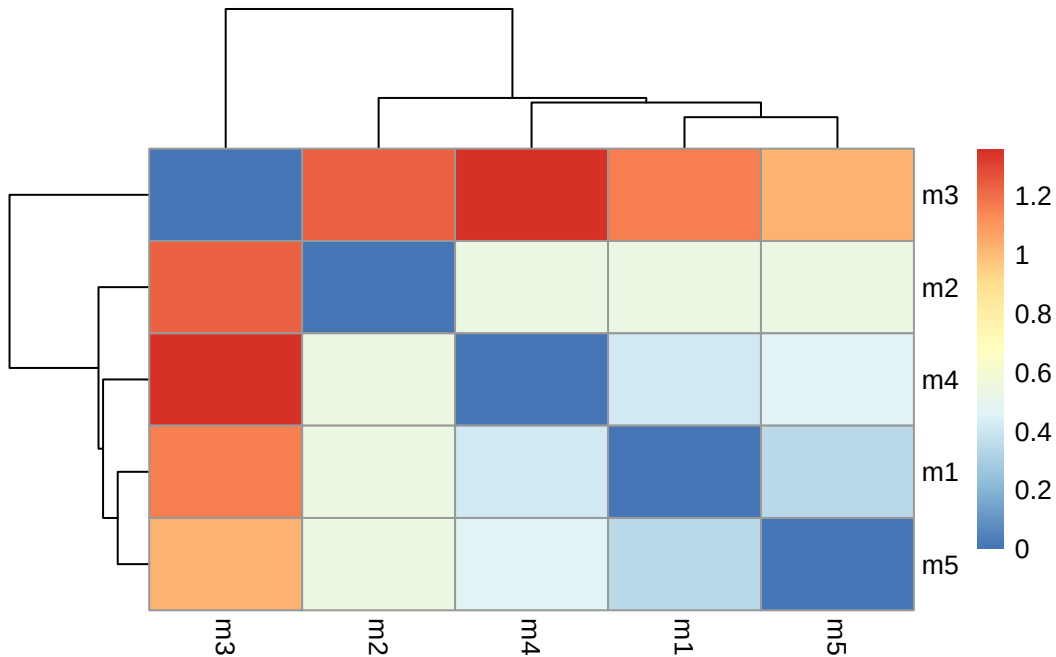
```
# Calculating pairwise RMSD between all models  
rd <- rmsd(pdbs, fit = TRUE)
```

Warning in rmsd(pdbs, fit = TRUE): No indices provided, using the 99 non NA positions

```
# Printing RMSD range  
range(rd)
```

```
[1] 0.00 1.36
```

```
# Rows and columns for labels  
colnames(rd) <- paste0("m", 1:5)  
rownames(rd) <- paste0("m", 1:5)  
  
# Drawing RMSD heatmap  
pheatmap(rd)
```



The RMSD heatmap compares structural similarity between the AlphaFold models. Lower RMSD values show more similar structures, while higher values indicate greater structural differences between models.

```
# Reading experimental HIV protease structure
pdb <- read.pdb("1hsg")
```

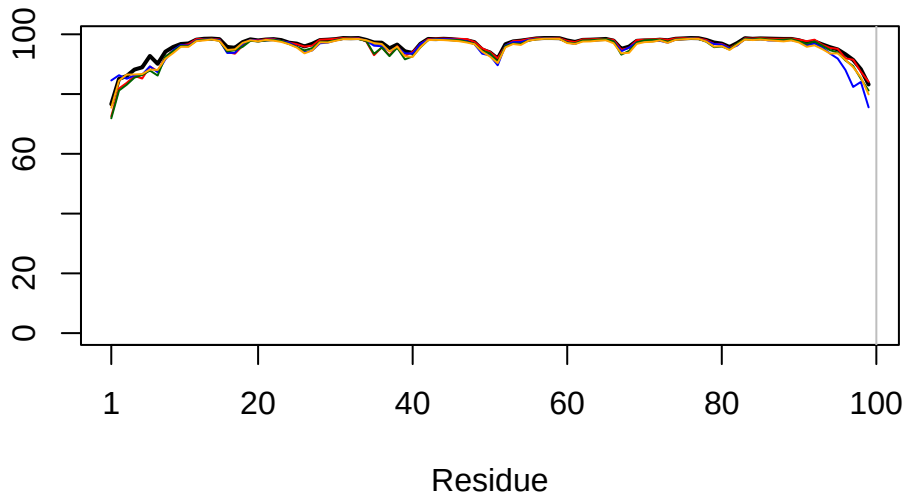
Note: Accessing on-line PDB file

```
# Plotting pLDDT confidence scores for all models
plotb3(pdb$b[1,],
  typ = "l",
  lwd = 2,
  sse = pdb)
```

Warning in plotb3(pdb\$b[1,], typ = "l", lwd = 2, sse = pdb): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
points(pdb$b[2,], typ = "l", col = "red")
points(pdb$b[3,], typ = "l", col = "blue")
points(pdb$b[4,], typ = "l", col = "darkgreen")
points(pdb$b[5,], typ = "l", col = "orange")
```

```
# Add divider between the two chains
abline(v = 100, col = "gray")
```



The pLDDT scores represent AlphaFold confidence values for each residue position. Most regions show high confidence values, showing reliable structure predictions. Lower confidence regions correspond to flexible or disordered regions.

```
# Find rigid core positions shared across all models
core <- core.find(pdbbs)
```

```
core size 98 of 99  vol = 4.205
core size 97 of 99  vol = 3.341
core size 96 of 99  vol = 2.657
core size 95 of 99  vol = 2.112
core size 94 of 99  vol = 1.644
core size 93 of 99  vol = 1.211
core size 92 of 99  vol = 0.859
core size 91 of 99  vol = 0.532
core size 90 of 99  vol = 0.314
FINISHED: Min vol ( 0.5 ) reached
```

```
# Print core residue positions
core.inds <- print(core, vol = 0.5)
```

```
# 91 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     3   3      1
2     7  96     90
```

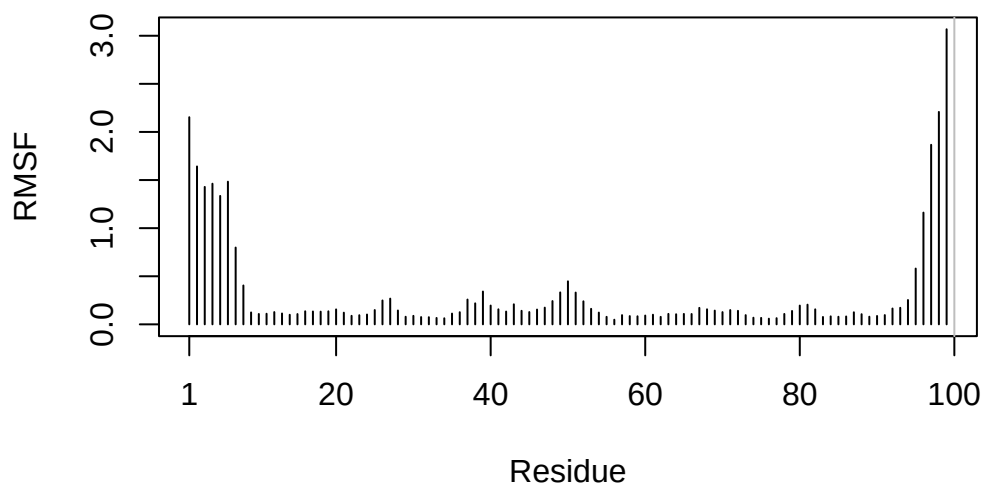
```
# Superpose structures using only rigid core residues
xyz <- pdbfit(pdb,
              core.inds,
              output = "corefit_structures")
```

```
# Calculate RMSF values
rf <- rmsf(xyz)

# Plot RMSF across residue positions
plotb3(rf,
       sse = pdb,
       ylab = "RMSF")
```

Warning in plotb3(rf, sse = pdb, ylab = "RMSF"): Length of input 'sse' does not equal the length of input 'x'; Ignoring 'sse'

```
# Add divider between chains
abline(v = 100, col = "gray")
```



The RMSF plot shows structural variability across residue positions. Higher RMSF values indicate regions with greater conformational variation between models.

```
# Find all JSON files containing PAE information
pae_files <- list.files(path = results_dir,
                        pattern = ".*model.*\\.json",
                        full.names = TRUE)
```

```
# Print JSON file names
basename(pae_files)
```

```
[1] "hivpr_monomer_94b5b_2_scores_rank_001_alphafold2_ptm_model_4_seed_000.json"
[2] "hivpr_monomer_94b5b_2_scores_rank_002_alphafold2_ptm_model_5_seed_000.json"
[3] "hivpr_monomer_94b5b_2_scores_rank_003_alphafold2_ptm_model_3_seed_000.json"
[4] "hivpr_monomer_94b5b_2_scores_rank_004_alphafold2_ptm_model_2_seed_000.json"
[5] "hivpr_monomer_94b5b_2_scores_rank_005_alphafold2_ptm_model_1_seed_000.json"
```

```
# Read PAE information for model 1 and model 5
pae1 <- read_json(pae_files[1],
                  simplifyVector = TRUE)

pae5 <- read_json(pae_files[5],
```

```
simplifyVector = TRUE)  
  
# Show stored information  
attributes(pae1)
```

```
$names  
[1] "plddt"    "max_pae" "pae"      "ptm"
```

```
# Print maximum PAE scores  
pae1$max_pae
```

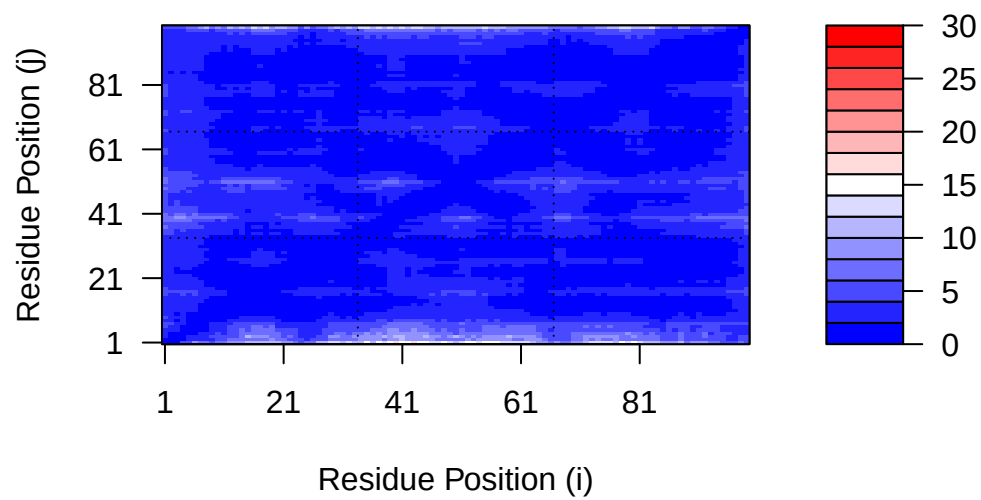
```
[1] 16.5625
```

```
pae5$max_pae
```

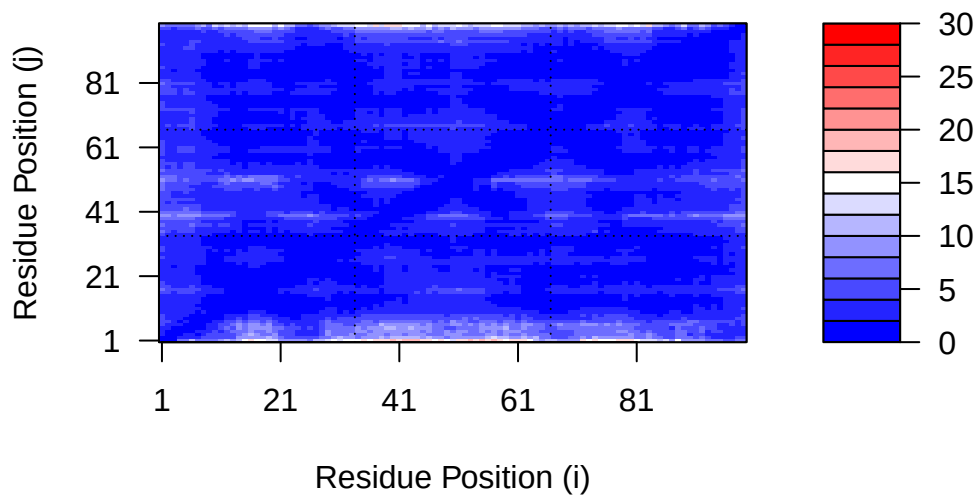
```
[1] 19.60938
```

Low maximum PAE values indicate better confidence in the relative positioning of residues and domains. Model 1 has a lower maximum PAE value than model 5, indicating somewhat that model 1 is more reliable overall.

```
# Plot PAE matrix for model 1  
plot.dmat(pae1$pae,  
          xlab = "Residue Position (i)",  
          ylab = "Residue Position (j)",  
          grid.col = "black",  
          zlim = c(0, 30))
```

```
# Plot PAE matrix for model 5
plot.dmat(pae5$pae,
  xlab = "Residue Position (i)",
  ylab = "Residue Position (j)",
  grid.col = "black",
  zlim = c(0, 30))
```



The PAE plots visualize AlphaFold confidence in the relative positioning between residues. Lower values indicate higher confidence. Model 1 shows generally lower PAE values, showing better predicted structural reliability than model 5.

```
# Find the multiple sequence alignment file
aln_file <- list.files(path = results_dir,
                      pattern = ".a3m$",
                      full.names = TRUE)
```

```
# Print alignment file name
aln_file
```

```
[1] "hivpr_dimer//hivpr_monomer_94b5b_2.a3m"
```

```
# Read the alignment file
aln <- read.fasta(aln_file[1],
                  to.upper = TRUE)
```

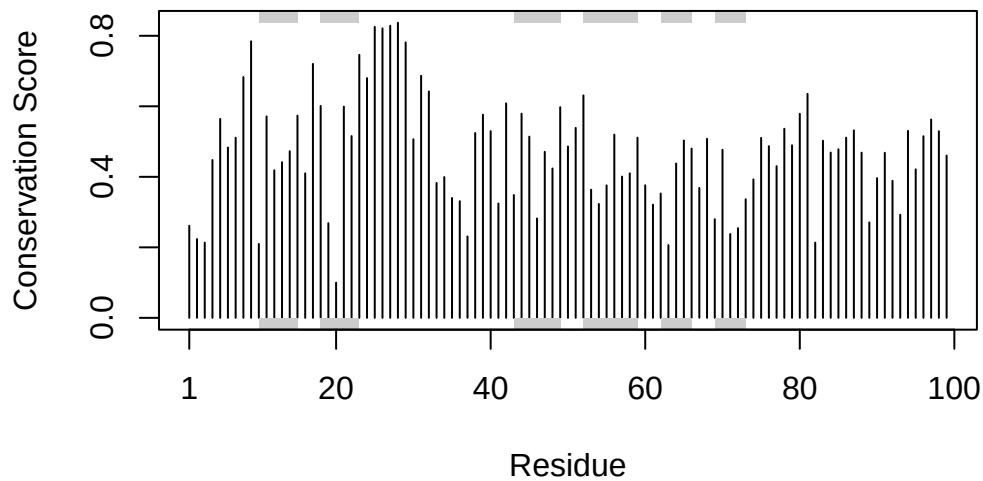
```
[1] " ** Duplicated sequence id's: 101 **"
```

```
# Check alignment dimensions
dim(aln$ali)
```

```
[1] 5397 132
```

```
# Calculate residue conservation scores
sim <- conserv(aln)
```

```
# Plot conservation scores for the first HIV protease chain
plotb3(sim[1:99],
      sse = trim.pdb(pdb, chain = "A"),
      ylab = "Conservation Score")
```



Highly conserved residues likely play important functional or structural roles in HIV protease. Conserved regions may correspond to catalytic residues or residues important for maintaining protein stability.

```
# Generate consensus sequence using a high conservation cutoff
con <- consensus(aln,
                 cutoff = 0.9)

# Print consensus sequence
con$seq
```

```

[1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
[37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"

```

The consensus sequence highlights highly conserved residues across the alignment. Strongly conserved positions are pretty functionally important.

```

# Read the top-ranked AlphaFold model
m1.pdb <- read.pdb(pdb_files[1])

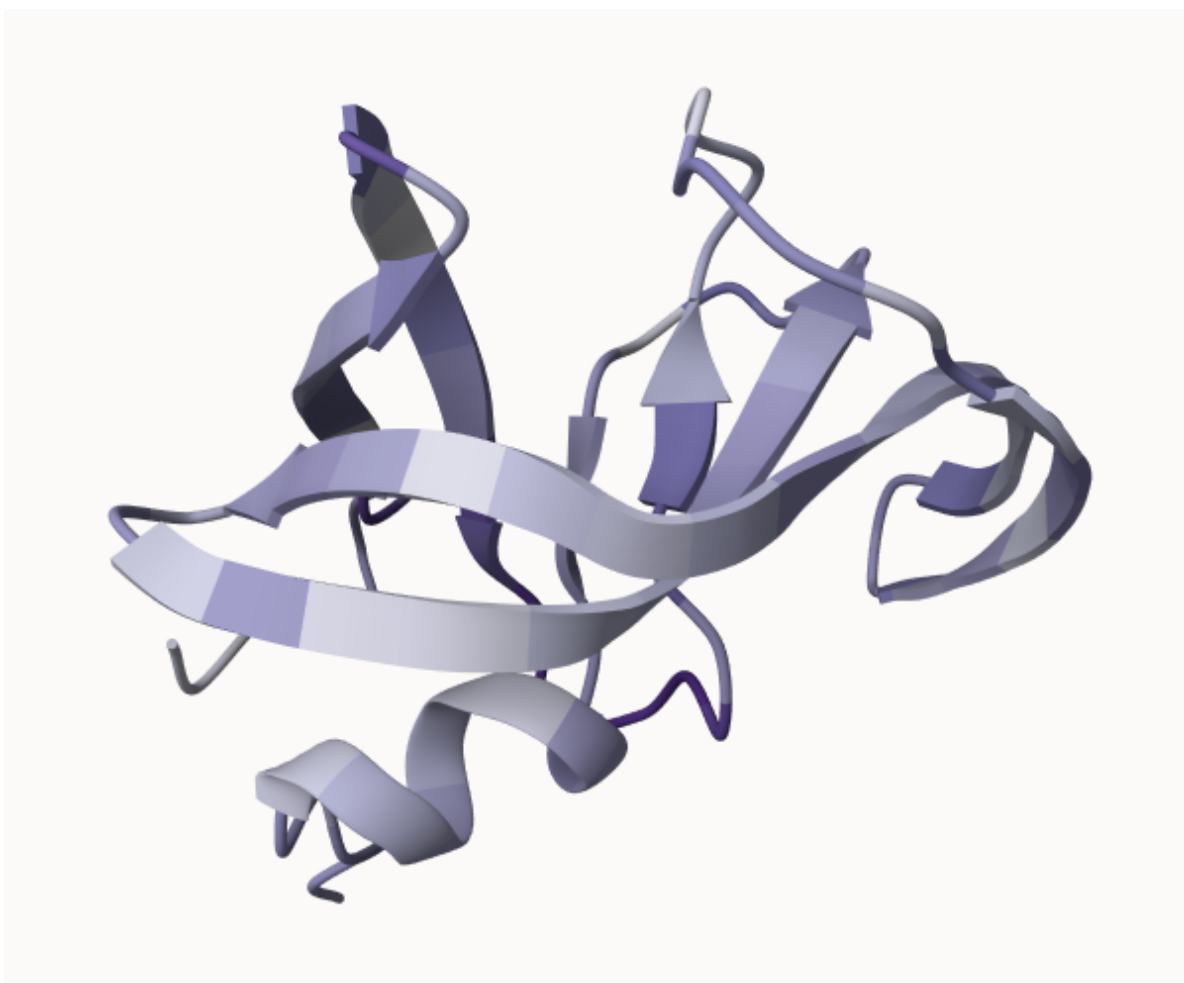
# Get the number of unique residues in the PDB model
n.res <- length(unique(m1.pdb$atom$resno))

# Use only the conservation scores that match the number of residues
sim.use <- sim[1:n.res]

# Map conservation scores onto the residue numbers in the PDB
occ <- vec2resno(sim.use, m1.pdb$atom$resno)

# Write a new PDB file with conservation scores in the occupancy column
write.pdb(m1.pdb,
          o = occ,
          file = "m1_conserv.pdb")

```



HIV protease AlphaFold model colored by residue conservation scores. Highly conserved active site residues are visible in the central region of the structure.

Summary

AlphaFold successfully predicted HIV protease structures with generally high confidence scores. RMSD and RMSF analyses showed structural similarities and variable regions among models. PAE analysis demonstrated that some models were more reliable than others. Conservation analysis identified highly conserved residues that are likely important for HIV protease function and catalytic activity.